

BCA-5003: Computer Network

Unit I — Detailed Study Guide, Corrected Notes and PYQ Focus

Scope: This guide is restricted to **Unit I: Basic Concepts, OSI Model and TCP/IP Model**. It consolidates the supplied Unit-I material and previous-year question patterns. The source text contains several OCR/technical inaccuracies; this guide uses corrected terminology and exam-appropriate explanations.

1. Unit I at a Glance

What you must prepare

1. Basics of computer networks and data communication.
2. Components and characteristics of data communication.
3. Communication modes and line configurations.
4. Network services and distributed processing.
5. Network topology: star, bus, ring, tree, mesh, and hybrid.
6. Error concepts, error detection, error correction, parity, CRC, and Hamming code basics.
7. OSI reference model and responsibilities of all seven layers.
8. TCP/IP reference model, comparison with OSI, and functions of TCP/IP.
9. MAC address versus IP address.
10. IPv4 and IPv6 fundamentals.
11. Network-layer responsibilities.
12. Virtual-circuit and datagram networks.
13. DTE–DCE interface and parallel transmission.

High-priority PYQ areas

Topic	Appeared in supplied PYQs
CRC and CRC numerical	2021, 2023–24, 2025–26
OSI layers and functions	2015, 2016, 2018, 2019, 2021
Network topology	2019, 2023–24
Data communication characteristics/components	2017, 2018, 2021, 2024–25
MAC vs IP address	2024–25

Topic	Appeared in supplied PYQs
IPv4/IPv6 and network-layer responsibilities	2024–25
TCP/IP functions	2025–26
Error detection/correction/Hamming code	2016–2021
Virtual circuit vs datagram	2023–24
DTE–DCE, line configuration, parallel transmission	2021–22

2. Computer Networks and Data Communication

2.1 Computer Network

A **computer network** is a collection of interconnected autonomous devices—such as computers, servers, phones, printers, switches, and routers—that communicate and share resources using agreed rules called protocols.

A network can use:

- **Wired media:** twisted-pair cable, coaxial cable, optical fibre.
- **Wireless media:** radio waves, Wi-Fi, microwave, cellular, satellite.

Why networks are used

- Share hardware such as printers and storage.
- Share files, databases, applications, and Internet access.
- Support email, messaging, web browsing, cloud platforms, video calls, and remote access.
- Allow centralized administration, backup, authentication, and security policies.
- Reduce cost by allowing common resources to be used by many users.
- Support scalability: new devices and services can be added as requirements grow.

Exam definition

A computer network is an interconnection of two or more autonomous computing devices through communication links for the purpose of sharing data, resources, and services.

2.2 Data Communication

Data communication is the exchange of data between two or more devices through a transmission medium. The information may be text, numbers, images, audio, video, or program data.

Data communication is successful only when the sender and receiver follow compatible rules for formatting, addressing, timing, transmission, and error handling.

Basic components of a data-communication system

Component	Meaning	Example
Message	Information to be communicated	File, email, image, audio, video
Sender / transmitter	Device that originates and sends the message	Laptop sending a file
Receiver	Device that accepts the message	Server receiving the file
Transmission medium	Physical or wireless path taken by signals	Fibre cable, Wi-Fi, radio link
Protocol	Set of rules governing communication	TCP/IP, HTTP, Ethernet

Important correction

A **protocol** is not merely an optional software feature. Communication between different devices is reliable only when they agree on protocol rules, including data format, timing, addressing, sequencing, and error control.

2.3 Characteristics of Effective Data Communication

A good data-communication system is evaluated using four principal characteristics.

1. Delivery

Data must reach the **correct destination**, intended device, process, or user. A packet delivered to the wrong host is not a successful delivery.

2. Accuracy

Data must arrive without unwanted alteration. Bit errors, missing bytes, duplication, or corrupted frames reduce accuracy. Error-detection and correction mechanisms help maintain accuracy.

3. Timeliness

Data must arrive when it is needed. For stored files, a small delay may be acceptable. For live audio, online gaming, remote control, and video conferencing, late data can be useless.

4. Jitter

Jitter is the variation in packet delay or packet-arrival time. It is especially harmful to real-time audio and video.

Example: if voice packets should arrive every 20 ms but some arrive after 20 ms and others after 70 ms, the sound may become choppy even if no packet is lost.

Jitter control

Common approaches include:

- Jitter buffers at the receiver to temporarily store early packets.
- QoS mechanisms to prioritize time-sensitive traffic.
- Sufficient bandwidth and congestion control.
- Stable routes and low-delay network paths.

Exam line: Jitter is the variation in delay between the arrivals of consecutive packets in a packet-switched network.

3. Communication Modes and Configurations

3.1 Direction of Data Flow

Mode	Description	Example
Simplex	Data travels in only one direction	Keyboard to computer; broadcast TV
Half-duplex	Both directions are possible, but only one direction at a time	Walkie-talkie
Full-duplex	Both sides can send and receive simultaneously	Telephone call; switched Ethernet

3.2 Line Configuration

A **line configuration** describes how devices are connected to a communication link.

Point-to-point configuration

A dedicated link connects exactly two devices. The complete capacity of the link is available to those two endpoints.

Examples: a direct router-to-router leased line, a serial cable between two devices, a point-to-point microwave link.

Advantages: simple communication, dedicated bandwidth, easier fault isolation.

Limitation: more links may be required when many devices must connect.

Multipoint configuration

More than two devices share a common physical link. Link capacity is shared among attached devices.

Sharing may be:

- **Spatial sharing:** multiple devices use separate portions of the medium at the same time.
- **Time sharing:** devices take turns using the channel.

Example: a shared bus network.

3.3 One-to-Many Communication

One sender distributes information to many receivers. Examples include television broadcasting, a teacher streaming a lecture, a multicast update, and a public web server serving many clients.

Do not confuse **one-to-many** with point-to-point. One-to-many describes the communication relationship; point-to-point describes a dedicated physical/logical link between two endpoints.

3.4 Serial and Parallel Transmission

Serial transmission

Bits are transmitted one after another over a single communication line or logical channel.

- Economical for long distances.
- Less affected by clock skew across wires.
- Used in modern high-speed interfaces and networking links.

Parallel transmission

Multiple bits are sent simultaneously over multiple separate lines. For an n-bit parallel transfer, n data lines are generally required, plus control/clock lines.

- Can transfer a group of bits simultaneously over short distances.
- Needs more wires, connectors, and synchronization.
- Becomes unreliable or costly over long distances because bits may arrive at slightly different times; this is called **skew**.

Exam point: Parallel transmission is usually suitable for short distances; serial transmission is generally preferred for long-distance communication.

4. Network Services and Distributed Processing

4.1 Major Network Services

Network services are facilities made available to users and applications through a network.

Directory services

A directory service stores and organizes information about users, devices, groups, and resources.

Functions include:

- User-account and password information.

- **Authentication:** verifying identity.
- **Authorization:** deciding which resources a verified user may access.
- **Mapping names to information,** such as a user name to an account record.

Domain Name System (DNS)

DNS translates human-readable domain names, such as `www.example.com`, into IP addresses used for network delivery.

File services

- **File sharing:** making files available to authorized users.
- **File transfer:** copying or moving files between systems.

Communication services

- Email.
- Instant messaging and chat.
- Voice/video conferencing.
- Discussion forums and collaboration platforms.
- Remote desktop and remote access.

Application services

- Web services.
- Database services.
- Print services.
- Storage services.
- Shared server applications.

4.2 Distributed Processing

Distributed processing is an arrangement in which processing work is divided among multiple computers or processors connected by a network. Each system performs part of the work and coordinates with the others.

Examples: cloud services, distributed databases, microservices, and MapReduce-style data processing.

Benefits

- Better performance through parallel work.
- Resource sharing.
- Improved reliability; one node's failure may not stop all work.

- Scalability by adding machines.
- Work can be placed near users or data.

Challenges

- Coordination and synchronization.
- Network delay and failures.
- Security across multiple systems.
- More difficult debugging and data consistency.

5. Network Topology

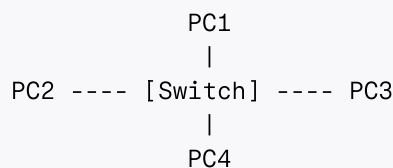
5.1 Meaning of Topology

Network topology is the physical or logical arrangement of network devices and the links between them. It determines how devices are connected and how data moves through the network.

- **Physical topology:** actual layout of cables, devices, ports, and links.
- **Logical topology:** path followed by data, which may differ from physical arrangement.

5.2 Star Topology

In a **star topology**, every node has an individual link to a central device, normally a switch. Communication between two nodes passes through the central device.



Advantages

- Failure of one node cable normally affects only that node.
- Easy installation, expansion, and fault isolation.
- Centralized management.
- Good performance with an Ethernet switch because each link can be dedicated and full-duplex.

Disadvantages

- Failure of the central switch/hub can disrupt the whole network.
- Requires more cable than bus topology.
- Central device adds equipment cost.

5.3 Bus Topology

All nodes share a single backbone cable. A signal sent on the bus is available along the shared medium; only the intended receiver accepts the frame.

```
[T] ---- PC1 ---- PC2 ---- PC3 ---- [T]
```

[T] represents a terminator, required in traditional coaxial bus networks to prevent signal reflection.

Advantages

- Uses less cable for small installations.
- Low initial cost.
- No central networking device is necessary in its basic form.

Disadvantages

- Break or fault in the backbone can affect the complete network.
- Difficult troubleshooting.
- Shared medium can suffer collisions and performance degradation.
- Limited cable length and number of nodes.

5.4 Ring Topology

In a **ring topology**, each node connects to two neighboring nodes, creating a closed loop. Data passes from node to node around the ring. Traditional token-ring designs use a token to control transmission.

```
PC1 ---- PC2  
|        |  
PC4 ---- PC3
```

Advantages

- Orderly access when token passing is used.
- Predictable performance under controlled conditions.
- Each node can receive an equal opportunity to transmit.

Disadvantages

- A cable or node failure can disrupt a simple ring.
- Adding/removing a device can interrupt the network.
- Delay may rise as more nodes are added.

5.5 Tree Topology

A **tree topology** has a hierarchical structure with a root or backbone and multiple branches. It can be viewed as interconnected star networks arranged in levels.

Advantages

- Highly scalable.
- Supports hierarchical administration.
- A local branch failure may be isolated from other branches.

Disadvantages

- Backbone/root device failure can affect large portions of the network.
- More cabling and active devices increase cost and complexity.
- Upper-level devices can become bottlenecks.

5.6 Mesh Topology

A **mesh topology** connects nodes with multiple point-to-point links.

- **Full mesh:** every node has a direct link to every other node.
- **Partial mesh:** only important nodes have multiple redundant links.

For a full mesh containing n devices, the number of links is:

$$\frac{n(n-1)}{2}$$

Advantages

- High reliability and fault tolerance due to alternative paths.
- High privacy because direct links can be used.
- A single link failure usually does not stop communication.

Disadvantages

- Very expensive for many devices.
- Difficult installation and management.
- Requires many ports, cables, and interfaces.

5.7 Hybrid Topology

A hybrid topology combines two or more basic topologies, such as star-bus or star-ring. Modern enterprise networks often use hybrid/hierarchical designs.

5.8 Topology Comparison

Topology	Central device	Main strength	Main weakness	Typical use
Star	Yes	Easy management and fault isolation	Central device is a single critical point	Modern LANs
Bus	No	Low cable cost for small legacy networks	Backbone failure affects all	Legacy coaxial Ethernet
Ring	No, in basic form	Controlled access	Break can interrupt ring	Legacy Token Ring/FDDI concepts
Tree	Yes, at multiple levels	Scalable hierarchy	Root/backbone dependency	Campus/enterprise LAN
Mesh	Not required	Very high reliability	Expensive and complex	WAN backbones, wireless mesh

6. Errors in Data Communication

6.1 Meaning of Error

An **error** occurs when the data received is different from the data sent. In binary communication, noise, interference, attenuation, synchronization problems, or equipment faults may change a 0 to 1 or a 1 to 0.

6.2 Types of Bit Errors

Single-bit error

Only one bit of a data unit changes.

Example:

```
Sent:      1001001
Received: 1000001
           ^ one bit changed
```

Multiple-bit error

More than one bit changes, but the changed bits need not be adjacent.

Burst error

Two or more bits within a span are affected. The first and last bits of the span are erroneous; bits in between may or may not be erroneous. Burst errors are common in real communication channels.

6.3 Error Detection versus Error Correction

Aspect	Error detection	Error correction
Purpose	Determines whether an error is present	Locates and fixes an error, or recovers correct data
Common techniques	Parity, checksum, CRC	Hamming code, forward error correction; retransmission-based ARQ
Extra data required	Redundant bits	Usually more redundancy
Receiver action	Reject/request retransmission if error found	Correct data directly if possible, or request retransmission

6.4 Parity Check

A **parity bit** is an additional bit added to a data unit to make the total number of 1s either even or odd.

Even parity

Set the parity bit so that the total number of 1s, including the parity bit, is even.

Example: Data = 1011001 has four 1s. For even parity, parity bit = 0.

Odd parity

Set the parity bit so that the total number of 1s, including parity, is odd.

Example: Data = 1011001 has four 1s. For odd parity, parity bit = 1.

Limitation of simple parity

Simple parity detects any odd number of changed bits, but it cannot detect an even number of changed bits. It also cannot identify the erroneous bit.

6.5 Checksum

In a checksum method, the sender divides data into fixed-size words, adds them using one's-complement arithmetic, and sends the complement of the sum as the checksum. The receiver repeats the calculation and checks whether the result is valid.

Checksum is used by several Internet protocols, but it is generally weaker than a well-designed CRC for detecting burst errors.

7. Cyclic Redundancy Check (CRC)

7.1 Definition

Cyclic Redundancy Check (CRC) is an error-detection technique based on modulo-2 binary division. The sender appends a calculated remainder, called the CRC or Frame Check Sequence (FCS), to the data. The receiver divides the received codeword by the same generator; a non-zero remainder indicates that an error has been detected.

Important correction

CRC does **not** prove that received data is absolutely error-free when the remainder is zero. It means that no error detectable by that CRC generator has been found. In exams, write: “The frame is accepted if the remainder is zero; otherwise it is considered erroneous.”

7.2 Terms used in CRC

Term	Meaning
Dataword	Original data bits supplied by upper layer
Generator / divisor	Predetermined bit pattern used for division
Degree of generator	Highest exponent of the generator polynomial; if generator length is $r+1$, degree is r
CRC remainder	r -bit remainder resulting from modulo-2 division
Codeword	Dataword followed by CRC bits

7.3 CRC Procedure at Sender

Suppose the generator has length $r + 1$. Then the CRC has r bits.

1. Choose a generator bit pattern G whose first and last bits are 1.
2. Append r zeros to the dataword.
3. Divide the augmented dataword by G using modulo-2 division, where subtraction is XOR.
4. Obtain an r -bit remainder R .
5. Replace the appended zeros with R .
6. Send dataword || R , called the codeword.

7.4 CRC Procedure at Receiver

1. Receive the complete codeword.
2. Divide it by the same generator using modulo-2 division.
3. If the remainder is all zeros, accept the frame.
4. If the remainder is non-zero, reject the frame and normally request retransmission at an appropriate protocol layer.

7.5 CRC Numerical Example

Question: Given dataword 1001 and divisor 1011, generate the CRC codeword.

Step 1: Find number of appended zeros

Divisor 1011 has 4 bits. Therefore:

$$r = 4 - 1 = 3$$

Append three zeros to the dataword:

```
Dataword:          1001
Augmented dataword: 1001000
Divisor:           1011
```

Step 2: Perform modulo-2 division

Modulo-2 subtraction is XOR:

$$1001 \text{ XOR } 1011 = 0010$$

Continuing the XOR division of 1001000 by 1011 produces the 3-bit remainder:

$$R = 110$$

Step 3: Form codeword

Replace the three appended zeros with the remainder:

```
Codeword = dataword + CRC
          = 1001 110
          = 1001110
```

Verification

$$1001110 \div 1011 \rightarrow \text{remainder } 000$$

Therefore, the transmitted CRC codeword is:

1001110

7.6 Why CRC is Preferred

- Very effective at detecting common burst errors.
- Efficient hardware implementation using shift registers and XOR logic.
- Used in many data-link technologies, including Ethernet frame checking.
- Stronger error-detection capability than simple parity.

8. Hamming Code Basics

8.1 Purpose

A Hamming code is an error-correcting code that adds multiple parity bits at positions that are powers of two: 1, 2, 4, 8, and so on. In its basic form, it can correct a single-bit error.

8.2 Number of Redundant Bits

If there are m data bits and r parity bits, choose r so that:

$$2^r \geq m + r + 1$$

The extra +1 represents the no-error condition.

8.3 Hamming (7,4) Positions

A seven-bit Hamming code uses parity bits at positions 1, 2, and 4:

Position	1	2	3	4	5	6	7
Role	P1	P2	D1	P4	D2	D3	D4

For even parity:

- P1 checks positions 1, 3, 5, 7.
- P2 checks positions 2, 3, 6, 7.
- P4 checks positions 4, 5, 6, 7.

At the receiver, failed parity checks form a binary number called the **syndrome**. Its decimal value identifies the error position.

PYQ method: received word 1011011 with even parity

Number bit positions from left to right as 1 to 7:

```
Position:  1 2 3 4 5 6 7
Received:  1 0 1 1 0 1 1
```

- Check P1 positions 1, 3, 5, 7: 1, 1, 0, 1 $\rightarrow$ three 1s $\rightarrow$ odd $\rightarrow$ P1 check fails.
- Check P2 positions 2, 3, 6, 7: 0, 1, 1, 1 $\rightarrow$ three 1s $\rightarrow$ odd $\rightarrow$ P2 check fails.

- Check P4 positions 4, 5, 6, 7: 1, 0, 1, 1 $\rightarrow$ three 1s $\rightarrow$ odd $\rightarrow$ P4 check fails.

Syndrome = $111_2 = 7$. Therefore, bit position 7 is erroneous.

Correct the received word by flipping bit 7:

```
Received word: 1011011
Correct word:  1011010
```

Note: Always state the bit-positioning convention used in a numerical answer. Some books number positions right-to-left; follow the convention specified in the question or clearly declare your convention.

9. Error Control in Data-Link Layer

The data-link layer provides **node-to-node delivery**. It detects and manages lost, damaged, or duplicate frames over a single link.

9.1 Major error-control tools

- Error-detection code: parity, checksum, CRC.
- Sequence numbers: identify frames and detect duplicates.
- Acknowledgement (ACK): confirms successful receipt.
- Negative acknowledgement (NAK): asks for retransmission, where used.
- Timers: detect a missing acknowledgement.
- Retransmission: sender sends the frame again.
- ARQ: Automatic Repeat reQuest protocols.

9.2 Stop-and-Wait ARQ

1. Sender transmits one frame.
2. Sender waits for an ACK.
3. If ACK arrives correctly, sender transmits the next frame.
4. If the timer expires or a NAK is received, sender retransmits the frame.

Advantage: simple.

Disadvantage: inefficient on long-delay or high-bandwidth links because the sender remains idle while waiting.

9.3 Go-Back-N ARQ

The sender can send multiple frames up to a window size before waiting. If frame k is lost/damaged, the receiver discards that frame and normally subsequent out-of-order frames. The sender retransmits frame k and all following frames in the window.

Advantage: better link utilization than stop-and-wait.

Disadvantage: may retransmit correctly received frames.

9.4 Selective Repeat ARQ

The receiver can accept and buffer correctly received out-of-order frames. The sender retransmits only the missing/damaged frame.

Advantage: more efficient retransmission.

Disadvantage: more complex sender/receiver logic and buffering.

10. OSI Reference Model

10.1 Introduction

The **Open Systems Interconnection (OSI) reference model** is a seven-layer conceptual framework developed by ISO to standardize the functions of network communication. It explains what each layer does, allowing systems with different internal designs to communicate through standard interfaces and protocols.

Layer order

```
7 Application
6 Presentation
5 Session
4 Transport
3 Network
2 Data Link
1 Physical
```

Mnemonic from Layer 7 to Layer 1:

All People Seem To Need Data Processing.

10.2 Encapsulation and PDUs

As data moves down the sender's stack, each layer adds control information. This process is called **encapsulation**. At the receiver, the added information is removed upward through **decapsulation**.

OSI layer	Common Protocol Data Unit (PDU)
Application, Presentation, Session	Data
Transport	Segment (TCP) / Datagram (UDP)
Network	Packet
Data Link	Frame
Physical	Bits

10.3 Physical Layer — Layer 1

The physical layer transmits raw bits over a physical medium.

Functions

- Defines electrical, optical, mechanical, and radio characteristics of interfaces and media.
- Represents bits as signals.
- Defines bit rate/data rate.
- Defines connectors, pin layouts, voltages, frequencies, modulation, and line coding.
- Defines physical topology.
- Defines transmission modes: simplex, half-duplex, and full-duplex.

Examples: cables, fibre, radio signals, repeaters, hubs, Ethernet physical standards.

10.4 Data-Link Layer — Layer 2

The data-link layer provides node-to-node delivery over a single network link. It transforms raw bits into manageable units called frames.

Functions

- **Framing:** groups bits into frames.
- **Physical/MAC addressing:** uses local hardware addresses such as Ethernet MAC addresses.
- **Error control:** detects damaged/lost frames and may request retransmission.
- **Flow control:** prevents a fast sender from overwhelming a slower receiver on the link.
- **Media access control:** decides which device may use a shared medium.

Examples: Ethernet, Wi-Fi MAC, switches, bridges, PPP.

10.5 Network Layer — Layer 3

The network layer provides source-to-destination delivery across multiple interconnected networks.

Functions

- **Logical addressing:** uses IP addresses.
- **Routing/path determination:** selects a suitable path from source network to destination network.
- **Forwarding:** moves packets to the next hop.
- **Internetworking:** enables communication across separate networks.
- **Fragmentation and reassembly:** may occur when packet size requirements differ, depending on protocol/version.
- May support congestion-related functions.

Examples: IPv4, IPv6, ICMP, routers.

10.6 Transport Layer — Layer 4

The transport layer provides **process-to-process** delivery. It ensures data reaches the appropriate application process, not merely the correct machine.

Functions

- Segmentation of application data and reassembly at destination.
- Port addressing and multiplexing/demultiplexing.
- Connection control: connection-oriented or connectionless communication.
- End-to-end flow control.
- End-to-end error control, acknowledgements, sequencing, and retransmission where supported.
- Congestion control in TCP.

Examples: TCP and UDP.

10.7 Session Layer — Layer 5

The session layer manages dialogs or sessions between applications.

Functions

- Establishes, maintains, and terminates sessions.
- Dialog control: coordinates who can transmit and when.
- Supports half-duplex or full-duplex dialogue management.
- Synchronization/checkpoints in long transfers so recovery can resume from a known point.

In real Internet protocol stacks, these functions are often implemented by applications or transport/session-related libraries rather than as a distinct layer.

10.8 Presentation Layer — Layer 6

The presentation layer handles the syntax and representation of data so unlike systems can interpret it consistently.

Functions

- Translation and format conversion.
- Character-code conversion, such as ASCII/Unicode handling.
- Encryption and decryption.
- Compression and decompression.
- Data serialization and representation.

Examples: JPEG/PNG format handling, TLS encryption-related representation functions, UTF-8 encoding.

10.9 Application Layer — Layer 7

The application layer provides network services directly to end-user applications.

Functions

- Web access.
- Email services.
- File transfer and remote file access.
- Name/directory services.
- Remote login and network management.
- Shared database and cloud application access.

Examples: HTTP/HTTPS, SMTP, IMAP/POP3, FTP/SFTP, DNS, SSH.

10.10 OSI Layer Summary Table

Layer	Name	Main responsibility	Address / identifier	Typical device/protocol
7	Application	Services to user applications	Application-specific names	HTTP, DNS, SMTP
6	Presentation	Translation, encryption, compression	Data format	UTF-8, JPEG, TLS-related functions
5	Session	Dialog/session management, synchronization	Session ID	Session services
4	Transport	Process-to-process delivery, reliability, ports	Port number	TCP, UDP
3	Network	Logical addressing, routing, packet forwarding	IP address	IP, router

Layer	Name	Main responsibility	Address / identifier	Typical device/protocol
2	Data Link	Framing, local delivery, MAC, link error control	MAC address	Ethernet, switch
1	Physical	Signals and bit transmission	None	Cable, fibre, hub/repeater

11. TCP/IP Reference Model

11.1 Introduction

The **TCP/IP protocol suite** is the practical family of protocols used by the Internet and most modern networks. Unlike the OSI model, which is a seven-layer reference framework, TCP/IP is commonly presented as a four-layer model.

TCP/IP Model	Approximate OSI Mapping
-----	-----
Application	Application + Presentation + Session
Transport	Transport
Internet	Network
Network Access / Link	Data Link + Physical

Some books show a five-layer Internet model by separating Data Link and Physical layers. In BCA answers, write the four-layer model unless the question specifically follows a five-layer version.

11.2 TCP/IP Layers

1. Application layer

Provides services used by applications, including web browsing, email, file transfer, remote access, and name resolution.

Protocols: HTTP/HTTPS, DNS, SMTP, FTP, SSH, DHCP.

2. Transport layer

Provides end-to-end, process-to-process data transfer.

- **TCP:** connection-oriented, reliable byte-stream delivery; sequencing, acknowledgements, retransmission, flow and congestion control.
- **UDP:** connectionless, low-overhead datagram service; no built-in guarantee of delivery, order, or duplicate protection.

3. Internet layer

Provides logical addressing and routing across networks.

Protocols: IPv4, IPv6, ICMP, IGMP-related functions.

4. Network-access / Link layer

Handles local-network access and sending frames over the actual medium.

Technologies: Ethernet, Wi-Fi, PPP, ARP/Neighbor Discovery-related local functions.

11.3 Functions of TCP/IP

A correct exam answer should distinguish the suite as a whole from TCP and IP individually.

1. **Interconnects heterogeneous networks:** allows devices on different local and wide-area networks to communicate.
2. **Provides logical addressing:** IP identifies source and destination interfaces using IPv4 or IPv6 addresses.
3. **Supports routing:** routers forward IP packets toward destination networks.
4. **Provides process delivery:** TCP/UDP port numbers direct data to the correct application process.
5. **Supports reliable transfer when TCP is used:** TCP sequences data, acknowledges receipt, retransmits losses, and delivers data in order.
6. **Supports fast best-effort transfer when UDP is used:** UDP is useful where low overhead and timeliness matter more than guaranteed delivery.
7. **Performs segmentation and reassembly:** upper-layer messages can be divided into transport segments and reconstructed at the destination.
8. **Provides flow and congestion control through TCP:** TCP controls sender rate to protect the receiver and respond to network congestion.
9. **Supports applications through application protocols:** HTTP/HTTPS, DNS, SMTP, and many other protocols operate over TCP/IP.
10. **Supports extensibility and interoperability:** common standards allow diverse hardware, operating systems, and networks to communicate.

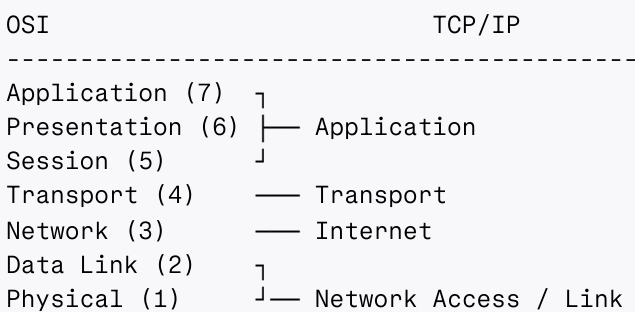
Important correction

- TCP performs error detection, ordered reliable delivery, and retransmission; it does not generally “correct” a corrupted packet in place.
- IP offers best-effort delivery. It does not guarantee delivery, order, or reliability.
- Security is not automatically provided by TCP/IP alone. Protocols such as TLS, IPsec, SSH, and VPN technologies may add security services.

12. OSI Model versus TCP/IP Model

Basis	OSI model	TCP/IP model
Nature	Reference model	Implemented protocol suite and model
Origin	ISO	U.S. DoD/DARPA Internet work
Number of layers	7	Usually 4
Upper layers	Separate Application, Presentation, Session	Combined Application layer
Lower layers	Separate Data Link and Physical	Combined Network Access/Link layer
Network layer service	Can describe connection-oriented or connectionless services	Internet layer is primarily connectionless IP
Real-world use	Teaching, standardization, conceptual design	Foundation of Internet and practical networking

Mapping diagram



13. MAC Address and IP Address

13.1 MAC Address

A **MAC (Media Access Control)** address is a link-layer hardware identifier used for delivery within a local network segment. In common Ethernet and Wi-Fi use, it is a 48-bit identifier written as six hexadecimal octets.

Example:

```
00:1A:2B:3C:4D:5E
```

A MAC address is typically assigned to a network interface by its manufacturer, but it can often be changed or spoofed in software. Therefore, do not write that MAC addresses “can never be changed.”

13.2 IP Address

An **IP address** is a logical network-layer address used to identify an interface and route packets between networks.

Examples:

```
IPv4: 192.168.1.10
IPv6: 2001:db8:1::10
```

IP addresses may be manually configured or dynamically assigned, usually by DHCP in IPv4 networks or by SLAAC/DHCPv6 in IPv6 networks.

13.3 Comparison

Feature	MAC address	IP address
Full form	Media Access Control address	Internet Protocol address
OSI layer	Data Link layer (Layer 2)	Network layer (Layer 3)
Purpose	Local delivery on a link/LAN	End-to-end routing across networks
Nature	Interface/hardware identifier	Logical, network-location-related identifier
Common length	Usually 48 bits (EUI-48)	IPv4: 32 bits; IPv6: 128 bits
Notation	Hexadecimal, e.g. 00:1A:2B:3C:4D:5E	IPv4 decimal dotted notation; IPv6 hexadecimal colon notation
Assignment	Usually vendor-assigned; can be locally administered/spoofed	Static/manual or dynamically assigned
Used by	Switches/bridges for local forwarding	Routers for packet forwarding
Scope	Local broadcast domain/link	Internetwork/global routing scope

13.4 How IP and MAC Work Together

When a device wants to send data to another device on the same IPv4 LAN, it knows the target IP address but needs the target MAC address for the Ethernet frame. It uses **ARP (Address Resolution Protocol)** to map IPv4 address to MAC address.

For IPv6, Neighbor Discovery Protocol performs a similar role; classic ARP is not used.

For a remote destination, the source normally sends the Ethernet frame to the MAC address of its default gateway, while the IP packet retains the remote destination's IP address. At each router hop, the Layer-2 frame/MAC addresses change, but the source and destination IP addresses normally remain end-to-end.

14. IPv4 Addressing

14.1 Definition

IPv4 (Internet Protocol version 4) uses a 32-bit logical address. It is commonly written in dotted-decimal notation as four octets.

Example:

192.168.1.1

The theoretical address space is:

$$2^{32} = 4,294,967,296$$

Not all addresses are available for ordinary public host assignment because some ranges are reserved for private use, multicast, loopback, documentation, and other special purposes.

14.2 IPv4 Address Structure

192.168.1.10
| | | |
8 bits each octet

An address has a network prefix and a host/interface part. Modern allocation uses **CIDR (Classless Inter-Domain Routing)** prefix notation.

Example:

192.168.1.0/24

Here /24 means the first 24 bits identify the network prefix.

14.3 Historical Classful Addressing

Classful addressing is historically important and may appear in university exams, but real networks now primarily use CIDR.

Class	First octet range	Default mask	Main historical use
A	1–126	/8	Very large networks
B	128–191	/16	Medium networks
C	192–223	/24	Small networks
D	224–239	Not applicable	Multicast
E	240–255	Not applicable	Experimental/reserved

Special notes:

- 127.0.0.0/8 is loopback.
- Private IPv4 ranges include 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

14.4 IPv4 Limitations

- Limited 32-bit address space.
- Address exhaustion in public IPv4 space.
- NAT is widely used to share public IPv4 addresses.
- Header is comparatively more complex than IPv6 header.
- Security is not inherent; IPsec support exists but deployment is optional.

15. IPv6 Addressing

15.1 Definition

IPv6 (Internet Protocol version 6) uses 128-bit addresses. It was designed to solve IPv4 address exhaustion and improve Internet-layer functionality.

The number of possible IPv6 addresses is:

$$2^{128}$$

This is an extremely large address space.

Example:

```
2001:0db8:0000:0000:0000:0000:0000:0001
```

Compressed form:

```
2001:db8::1
```

15.2 IPv6 Notation Rules

- An IPv6 address consists of eight groups of four hexadecimal digits.
- Leading zeros inside a group may be omitted.
- One consecutive run of all-zero groups may be compressed using ::.
- :: may be used only once in one IPv6 address to avoid ambiguity.

Example:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

15.3 IPv6 Features

- 128-bit address space.
- Simplified fixed base header.
- Extension headers for optional functions.
- Stateless Address Autoconfiguration (SLAAC) support.
- No broadcast; uses multicast and anycast instead.
- Better support for hierarchical routing.
- IPsec support is standardized for IPv6, but encryption is **not automatically mandatory for every IPv6 connection**.

15.4 IPv4 versus IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Address notation	Dotted decimal	Colon-separated hexadecimal
Example	192.168.1.1	2001:db8::1
Address space	About 4.29 billion theoretical addresses	2^{128} addresses
Header	Variable length, more fields	Simplified fixed base header
Broadcast	Supported	Not used; multicast/anycast used
NAT	Widely used due to scarcity	Usually not required for address conservation
Configuration	Manual or DHCP	SLAAC, DHCPv6, manual
Fragmentation	Routers and hosts may fragment	Source host handles fragmentation; routers do not fragment packets

16. Network Layer Responsibilities

The **network layer** is responsible for delivering packets from a source network to a destination network, possibly through many intermediate networks.

Core responsibilities

1. **Logical addressing:** adds source and destination IP addresses.
2. **Routing:** determines an appropriate path through interconnected networks.
3. **Forwarding:** sends each packet to the appropriate next-hop router/interface.
4. **Internetworking:** enables different networks and link technologies to communicate.

5. **Packet handling:** carries packets across networks and may support fragmentation/reassembly according to protocol rules.
6. **Traffic/congestion-related support:** network devices may queue, prioritize, drop, or signal packets when congestion occurs.

Very important distinction

- A **router** uses Layer-3 IP information to forward packets between networks.
- A **switch** normally uses Layer-2 MAC information to forward frames within a LAN.

17. Virtual Circuits and Datagram Networks

17.1 Virtual-Circuit Network

A **virtual circuit** is a logical, connection-oriented path established before data transfer. Once set up, packets follow the same logical path and typically carry a short virtual-circuit identifier rather than complete routing information.

Phases:

1. Connection setup.
2. Data transfer.
3. Connection release.

Examples: X.25, Frame Relay, ATM.

17.2 Datagram Network

A **datagram network** is connectionless. Each packet is independently routed and carries sufficient destination addressing information. Different packets belonging to the same message can take different paths and may arrive out of order.

Example: Internet Protocol (IP).

17.3 Comparison

Factor	Virtual circuit	Datagram
Connection setup	Required before data transfer	Not required
Service type	Connection-oriented	Connectionless
Route	Usually established once and followed by all packets	Chosen independently for each packet
Packet header	Usually shorter VC identifier after setup	Contains full destination addressing/routing information
Packet order	Usually preserved because packets follow same path	May be out of order

Factor	Virtual circuit	Datagram
Router state	Routers/switches maintain connection state	Routers normally do not maintain per-flow state for basic forwarding
Failure response	Established circuit may need to be recreated	Later packets can be routed through another path
Example	X.25, Frame Relay, ATM	IP

18. DTE and DCE Interface

18.1 DTE

DTE (Data Terminal Equipment) is a device that acts as the source or destination of user data.

Examples: computer, terminal, router interface in some serial WAN arrangements.

18.2 DCE

DCE (Data Circuit-terminating Equipment) is a device that provides the interface between DTE and the transmission medium. It may perform signal conversion, timing/clocking, and line access.

Examples: modem, CSU/DSU, or provider-side communication equipment.

18.3 Working

Computer (DTE) → Modem (DCE) → Transmission medium → Modem (DCE) → Computer (DTE)

At the sending side, DCE adapts DTE's data into signals suitable for the medium. At the receiving side, DCE converts received signals into a form usable by DTE.

Short comparison

DTE	DCE
Produces or consumes user data	Connects DTE to the transmission medium
Example: computer/terminal	Example: modem
Usually at customer/user end	Often provides line signaling/clocking functions

19. PYQ Answer Bank

Q1. Define data communication. State its basic components.

Answer: Data communication is the exchange of data between two or more devices through a transmission medium. A data-communication system has five basic components: message, sender, receiver, transmission medium, and protocol. The message is the information being transferred; the sender transmits it; the receiver accepts it; the medium carries signals; and the protocol defines the rules for communication.

Q2. State the characteristics of effective data communication.

Answer: Effective data communication has four characteristics: delivery, accuracy, timeliness, and jitter. Delivery means data reaches the correct destination. Accuracy means data arrives without error. Timeliness means data arrives when required, especially for real-time communication. Jitter is the variation in packet-arrival delay and can degrade voice/video quality.

Q3. What is a protocol? Why is it necessary?

Answer: A protocol is a set of rules that governs communication between devices. It specifies data format, addressing, timing, sequencing, error handling, and interpretation of messages. It is necessary because devices with different hardware and software cannot communicate meaningfully unless they follow common rules.

Q4. What is CRC? Explain its working.

Answer: CRC is an error-detection method based on modulo-2 division. The sender appends zeros to the dataword, divides the augmented dataword by a generator, and appends the obtained remainder as CRC bits. The resulting codeword is transmitted. The receiver divides the received codeword by the same generator. A zero remainder indicates that no detectable error is found; a non-zero remainder indicates an error and the frame is rejected.

Q5. Explain the functions of TCP/IP protocol suite.

Answer: TCP/IP enables communication across interconnected networks. IP provides logical addressing and routing. TCP provides reliable, ordered, connection-oriented delivery using sequence numbers, acknowledgements, retransmissions, flow control, and congestion control. UDP provides fast connectionless delivery when low overhead is preferred. TCP/IP also supports segmentation/reassembly, port-based process delivery, and applications such as web, email, DNS, and file transfer through application-layer protocols.

Q6. Draw OSI model and explain any three functions of each layer.

For a full answer, draw the seven layers in order and write at least three functions per layer from Section 10. In the exam, underline key terms: framing, MAC addressing, routing, logical addressing, segmentation, ports, encryption, session synchronization, and application services.

Q7. Differentiate MAC address and IP address.

Write the comparison table from Section 13.3. Mention clearly that MAC is used for local link delivery by switches, whereas IP is used for routing across networks by routers.

Q8. Write short notes on IPv4 and IPv6.

For IPv4: define 32-bit dotted-decimal addressing, explain prefix/network-host concept, address shortage, NAT, and CIDR/classful history.

For IPv6: define 128-bit hexadecimal colon notation, address compression, large address space, simplified header, SLAAC, multicast instead of broadcast, and improved route aggregation.

Q9. What are responsibilities of network layer?

The network layer provides logical addressing, routing, forwarding, internetworking, packet delivery across multiple networks, and packet handling such as fragmentation where required by protocol rules.

Q10. Compare virtual circuits and datagrams.

Use the table in Section 17.3. Start with the definitions: virtual circuit is connection-oriented with setup; datagram is connectionless with independent routing of packets.

Q11. Explain DTE–DCE interface.

DTE is the source/destination of user data, such as a computer. DCE interfaces DTE with the transmission medium, such as a modem. DCE converts data/signals into a form appropriate for the communication link and may supply timing/clocking.

Q12. Explain parallel transmission.

Parallel transmission sends multiple bits simultaneously over multiple wires. It is fast over short distances but requires more wires and suffers from timing skew over longer distances. Therefore, serial transmission is preferred for long distances.

20. Exam Writing Strategy

For 2-mark questions

- Give one exact definition.
- Add one key property or example.
- Do not add unrelated long explanation.

Example: Define jitter.

Jitter is the variation in packet-arrival delay in a network. It affects real-time applications such as voice and video conferencing by causing uneven playback.

For 5-mark questions

- Begin with definition.
- Draw a small labeled diagram if relevant.
- Write 4–6 numbered points.
- End with one use, advantage, limitation, or example.

For 10-mark questions

- Definition/introduction.
- Labeled diagram or table.
- Detailed explanation under subheadings.
- Advantages/disadvantages or comparison.
- Concluding practical example.

Diagrams to practice

1. Five components of data communication.
2. CRC sender and receiver flow.
3. Star, bus, ring, tree, and mesh topology.
4. Seven-layer OSI model.
5. OSI-to-TCP/IP mapping.
6. DTE–DCE connection.
7. Hamming (7,4) bit positions.

21. One-Day Revision Checklist

- [] Learn definitions: network, data communication, protocol, topology, jitter, CRC, DTE, DCE.
- [] Memorize data-communication components and four effectiveness characteristics.
- [] Practice topology diagrams with two advantages and two disadvantages each.
- [] Learn CRC algorithm and solve 1001 with divisor 1011.
- [] Memorize OSI layer order, PDU names, functions, and typical protocols/devices.
- [] Learn TCP/IP four layers and OSI mapping.
- [] Prepare MAC vs IP and IPv4 vs IPv6 comparison tables.
- [] Revise network-layer functions.
- [] Practice virtual-circuit vs datagram comparison.
- [] Revise Stop-and-Wait, Go-Back-N, and Selective Repeat ARQ.

22. Most Expected Questions

1. Explain CRC and show CRC-codeword generation for a given dataword and divisor.
2. Draw the OSI model and explain functions of each layer.
3. Explain the TCP/IP model. Compare it with the OSI model.
4. Define network topology. Explain star, bus, ring, tree, and mesh topology with merits and demerits.
5. Define data communication. Explain its components and characteristics.
6. Differentiate MAC address and IP address.
7. Write detailed notes on IPv4 and IPv6 addressing.
8. Explain the functions/responsibilities of the network layer.
9. Discuss error detection and correction techniques. Explain parity, CRC, and Hamming-code basics.
10. Compare virtual-circuit networks with datagram networks.
11. Explain DTE–DCE interface and line configurations.
12. Explain serial and parallel transmission.

Final Accuracy Notes

- CRC is based on **modulo-2 division/XOR**, not ordinary arithmetic division.
- CRC zero remainder means **no detectable error**, not an absolute proof that data is perfect.
- TCP offers reliable delivery; UDP does not guarantee delivery or ordering.
- IP routing works at the **network layer**; MAC-based frame forwarding works at the **data-link layer**.
- ARP maps an **IPv4 address** to a **MAC address** on a local network. IPv6 uses Neighbor Discovery rather than ARP.
- MAC addresses may be vendor-assigned but can often be locally administered or spoofed; they are not permanently unchangeable in all practical cases.
- IPv6 does not use broadcast, and IPsec support does not mean every IPv6 packet is automatically encrypted.

Prepared as a corrected, exam-oriented Unit-I guide for BCA-5003 Computer Network.